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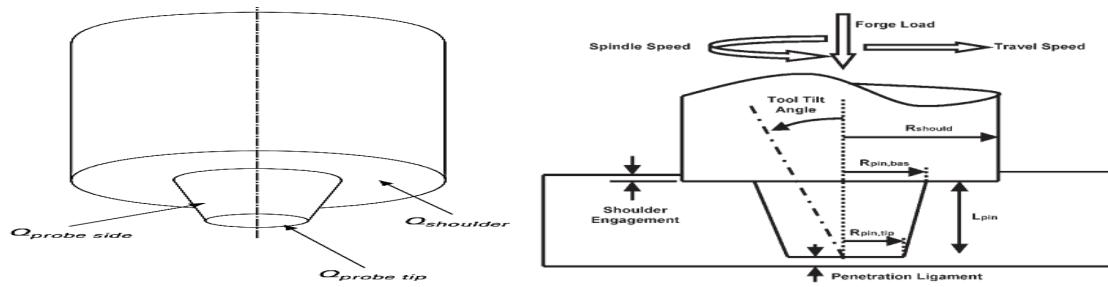


Fig. 1 Schematic tool geometry of heat generation contributions from the shoulder, probe side and probe tip.

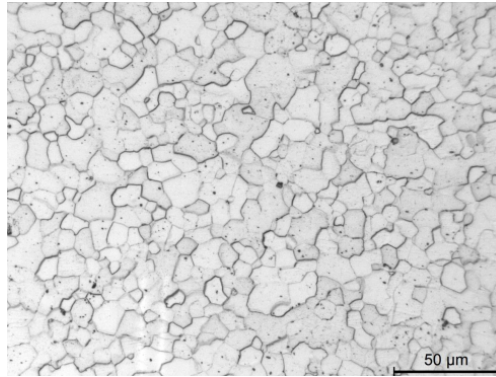


Fig. 2 Optical micrograph of IF steel sheet material

List of tables

Table 1 Chemical composition of the selected IF steel (in wt %)

C	Mn	S	Si	Al	N ₂	Ti	P	Fe
0.002	0.079	0.0089	0.006	0.0359	0.003	0.0657	0.0109	Balance

Table 2 Typical mechanical properties of IF steel at room temperature

Material properties	IF Steel
Yield stress (N/mm ²)	137
Density (kg/m ³)	7870
Thermal conductivity (W/m°C)	51.9
Specific heat (J/kg °C)	461
Young's modulus E (GPa)	200
Melting temperature (K)	1800
Frictional coefficient	0.4

Range of strain rates (per Second)	0.001 to 750
Tensile strength (N/mm ²)	302

Table 3 Heat generated during friction stir welding at Various Speed.

Rotation Speed (rpm)	Q _{shoulder-rotation}	Q _{shoulder-translation}	Q _{pin-rotation}	Q _{pin-translation}	Q _{pin tip-rotation}	Q _{pin tip-translation}	Total heat (watt)
250	684.366	179.398	32.438	63.459	13.8649	15.187	988.7
300	821.228	179.398	38.925	63.459	16.637	15.187	1134.8
350	957.228	179.398	45.372	63.459	19.639	15.187	1280.04
400	1094.98	179.398	51.9	63.459	22.184	15.187	1427.108
450	1231.843	179.398	58.388	63.459	24.956	15.187	1573.23
500	1395.04	179.398	66.123	63.459	28.263	15.187	1747.47
550	1505.594	179.398	71.363	63.459	30.5	15.187	1865.5
600	1642.457	179.398	77.850	63.459	33.275	15.187	2011.63
650	1779.346	179.398	84.339	63.459	36.048	15.187	2157.77
700	1916.21	179.398	90.826	63.459	38.82	15.187	2303.899

Table 4 Total torque and Power generated Friction stir Welding in Various speed and different diameter (Tool shoulder and pin dia.) by using design of Experiment.

SL. No.	Rotational Speed (rpm)	Shoulder dia. (mm)	Pin dia. (mm)	Torque (kN-mm)	Power (kW)	Local heat flux (W/mm ²)
1	250	18	6.0	127.4	3.34	12.559
2	250	16	5.5	90.413	2.365	11.265
3	250	15	5.0	74.5	1.949	10.443
4	250	12	4.0	38.75	1.014	8.357
5	350	18	5.5	126.344	4.628	17.186
6	350	16	6.0	91.484	3.351	16.084
7	350	15	4.0	72.835	2.668	13.896
8	350	12	5.0	39.9	1.46	12.251
9	450	18	5.0	124.866	5.88	21.539
10	450	16	4.0	87.758	4.133	18.807
11	450	15	6.0	76.56	3.606	19.739
12	450	12	5.5	41.4	1.949	16.433
13	600	18	4.0	123.69	7.767	27.587

14	600	16	5.0	88.935	5.585	26.181
15	600	15	5.5	75.49	4.74	25.692
16	600	12	6.0	42.472	2.667	22.542

Table 5 Analysis of variance table for torque

Analysis of variance for torque: (Response Surface Regression: R-sq = 98.32% R-sq(adj) = 97.90%, R-sq(pred) = 96.77%, S = 4.57174.					
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	14653.0	4884.3	233.69	0.000
Linear	3	14653.0	4884.3	233.69	0.000
Tool rotation speed	1	0.0	0.0	0.0	0.975
Shoulder dia.	1	14622.4	14622.4	699.61	0.000
Pin dia.	1	30.6	30.6	1.46	0.250
Residual Error	12	250.8	20.9		
Total	15	14903.8			

Table 6 Analysis of variance table for power

Analysis of variance for power: (Response Surface Regression: S = 0.367969, R-sq = 96.70%, R-sq(adj) = 95.88%, R-sq(pred) = 92.81%.					
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	47.6152	15.8717	117.22	0.000
Linear	3	47.6152	15.8717	117.22	0.000
Tool rotation speed	1	19.9632	19.9632	147.44	0.000
Shoulder dia.	1	26.6811	26.6811	197.05	0.000
Pin dia.	1	0.9709	0.9709	7.17	0.020
Residual Error	12	1.6248	0.1354		
Total	15	49.2401			

Table 7 Analysis of variance table for local heat flux

Analysis of variance for local heat flux: (Response Surface Regression: S = 0.439876, R-sq = 99.56%, R-sq(adj) = 99.46%, R-sq(pred) = 99.04%.					
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	530.799	176.933	914.42	0.000

Linear	3	530.799	176.933	914.42	0.000
Tool rotation speed	1	481.990	481.990	2491.02	0.000
Shoulder dia.	1	48.097	48.097	248.58	0.000
Pin dia.	1	0.711	0.711	3.68	0.079
Residual Error	12	2.322	0.193		
Total	15	533.121			

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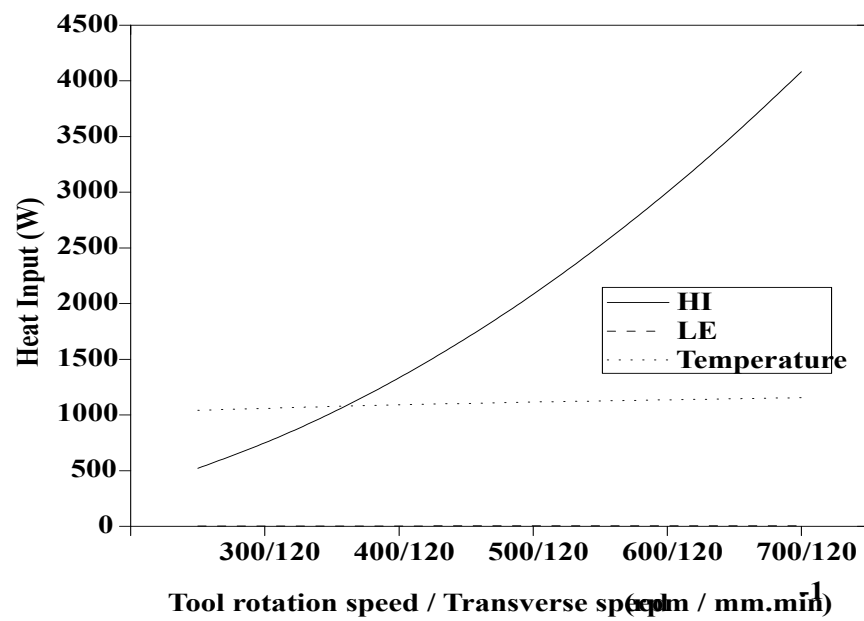


Fig. 3 Variation of Heat input with Tool rotation speed

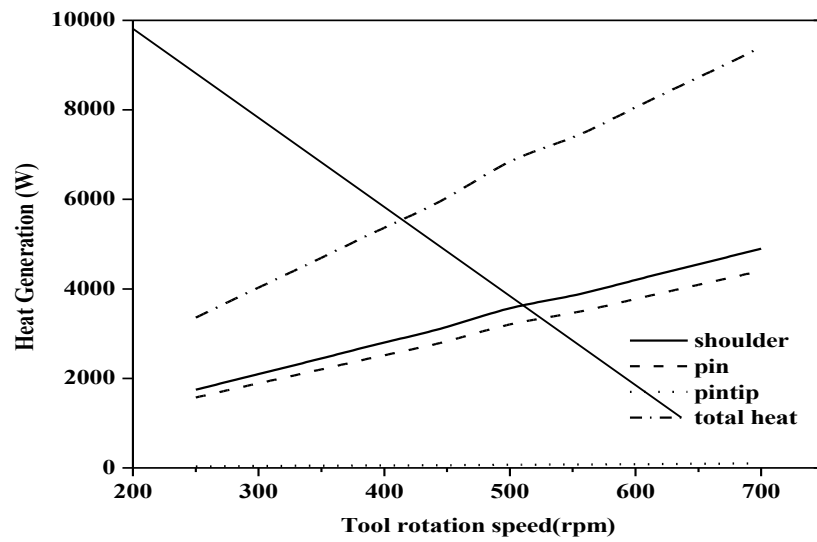


Fig. 4 Variation of heat generation with Tool rotation speed for tapered cylindrical pin

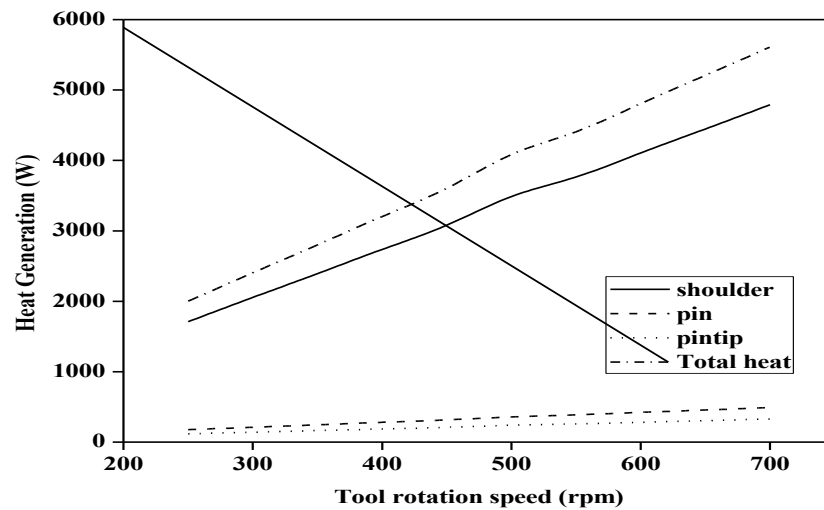


Fig. 5 Variation of heat generation with tool rotation speed for cylindrical pin profile

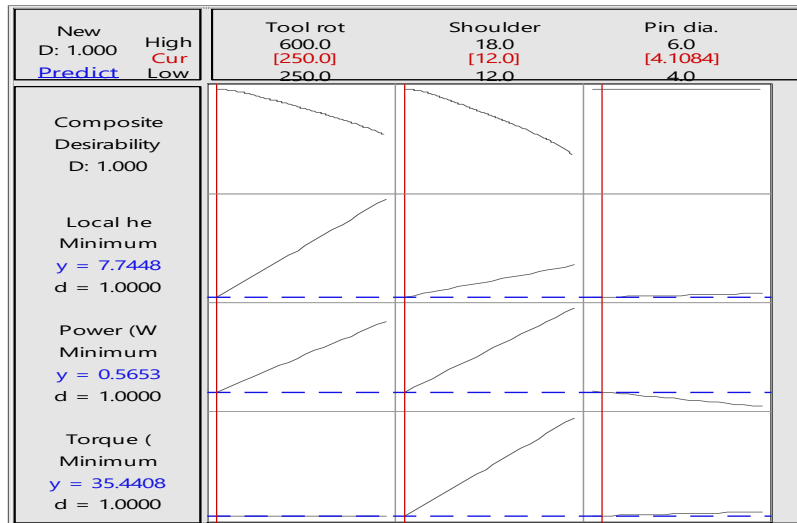


Fig. 6 Variation of torque, power and local heat flux against rotation speed, shoulder diameter and pin diameter.